

Use this template to begin typing the first page (summary page) of your electronic report. This template uses a 12-point Times New Roman font. Submit your paper as an Adobe PDF electronic file (e.g. 1111111.pdf), typed in English, with a readable font of at least 12-point type.

Do not include the name of your school, advisor, or team members on this or any page.

Be sure to change the control number and problem choice above.

You may delete these instructions as you begin to type your report here.

Follow us @COMAPMath on X or COMAPCHINAOFFICIAL on Weibo for the most up to date contest information.

$$P_{\text{total}} = \frac{dE}{dt} \quad (1)$$

$$P_{\text{total}} = P_{\text{screen}} + P_{\text{processor}} + P_{\text{network}} + P_{\text{sound}} + P_{\text{peripherals}} + P_{\text{battery loss}} \quad (2)$$

The goal is to solve

$$(\text{SOC})_{t,N} = \left(\frac{(\text{Energy left in battery})}{C_N} \right) \times 100\% = \left(\frac{E_0 - \int_0^t P_{\text{total}} dt}{C_N} \right) \times 100\% \quad (3)$$

and

$$t_{\text{remain}} = \frac{E_0 - \int_0^t P_{\text{total}} dt}{t_0^{-1} \int_0^{t_0} P_{\text{total}} dt}. \quad (4)$$

1 Screen

1.1 OLED

$$P = \eta \cdot J \cdot V \quad (5)$$

where η is the efficiency, J is the current density, and V is the voltage.

By Arrhenius equation, the carrier mobility, or the number of carriers whose energy exceeds the activation energy, is given by

$$\mu(T) = \mu_0 \exp -\frac{E_a}{kT} \quad (6)$$

where μ_0 is a constant, E_a is the activation energy, k is the Boltzmann constant, and T is the temperature in Kelvin.

Thus, stable illumination when P , J are constants

$$V \propto V_0 \exp \frac{E_a}{kT} \quad (7)$$

Now we may assume that the thermally-stimulated active migration rate k_{nr} is proportional to $\mu(T)$, i.e.

$$k_{nr} = k_0 \exp -\frac{E_a}{kT} \quad (8)$$

Lighting efficiency under high temperature

$$\eta = \frac{k_r}{k_r + k_{nr}} = \frac{1}{1 + A \exp -E_a/(kT)}, \quad A = \frac{k_0}{k_r} \quad (9)$$

where k_r is the rate of electromagnetic radiation. Here we assume P , V are constants, so

$$J \propto \eta^{-1} \propto 1 + A \exp -E_a/(kT) \quad (10)$$

hence,

$$P(T) \propto \exp \frac{E_{a1}}{kT} \left(1 + A \exp -\frac{E_{a2}}{kT} \right) \quad (11)$$

1.2 Influence of Ambient Lighting

Citing the works from previous scholars, we have

$$P_{\text{screen}} = a_0 + (Br)(a_R P_R + a_G P_G + a_B P_B) \quad (12)$$

where indices R, G, B denote red, green, and blue respectively; a_0 , a_R , a_G , a_B are constants related to the specific screen; P_R , P_G , P_B are the percentages of red, green, and blue respectively; and Br is the brightness of the screen.

We may assume that the user will adjust the brightness of the screen according to the ambient lighting L_{amb} , that the brightness of the screen is proportional to L_{amb} by a factor of ε , and that L_{amb} is roughly a trigonometric function of time t in a day, i.e.

$$Br = \varepsilon L_{\text{amb}} = \varepsilon L_0 \sin(\omega t + \delta) + L_{\text{base}} \quad (13)$$

where L_0 is the maximum ambient lighting in a day, δ is a phase shift depending on the geographical location, and L_{base} is the base ambient lighting (e.g. indoor lighting).

2 Network

Citing the works from previous scholars, we have the following equation

$$P_{\text{network}} = \sum_i \alpha_{ui} \theta_{ui} + \alpha_{di} \theta_{di} + \beta \quad (14)$$

where α_{ui} , α_{di} , β are constants related to the specific network type, and θ_{ui} , θ_{di} are the upload and download throughputs (in Mbps) respectively. The index i denote different processes.

3 Processing Units

3.1 CPU

$$P_{\text{CPU}} = P_{\text{static}} + P_{\text{dynamic}} \quad (15)$$

For every transistor, we consider its effect of storing charges, and thus see it as a tiny capacitor.

$$E_{\text{unit}} = \frac{1}{2} C U^2 \quad (16)$$

Here we consider the frequency of the CPU f , thus in a small cycle the energy consumed is E_{unit} . Thus, in unit time, the dynamic power consumption is proportional to $U^2 \cdot f$, i.e.

$$P_{\text{dynamic}} = \lambda U^2 \cdot f \quad (17)$$

where f is related to U . Suppose

$$f = b U^\mu \quad (18)$$

and the CPU utility is α , then

$$P_{\text{dynamic}} = \alpha U^{2+\mu} b \lambda \quad (19)$$

Citing the works from previous scholars, we have the static power consumption

$$P_{\text{static}} = U \left(c_1 T^2 e^{c_2/T} + I_{\text{gate}} \right) \quad (20)$$

where c_1 , c_2 are constants related to the specific CPU, and I_{gate} is the leakage current of the transistor gates.

3.2 GPU

Similar to CPU, we have

$$P_{\text{GPU}} = P_{\text{static}} + P_{\text{dynamic}} \quad (21)$$

$$P_{\text{dynamic}} = \alpha' U_{\text{GPU}}^{2+\mu'} b' \lambda' \quad (22)$$

$$P_{\text{static}} = U_{\text{GPU}} \left(c_3 T^2 e^{c_4/T} + I'_{\text{gate}} \right) \quad (23)$$

4 Speakers

Citing the works from previous scholars, we have

$$U_{\text{in,req}}(L_c) = 10^{\Delta L/20} U_{\text{ref}} \quad (24)$$

where $U_{\text{in,req}}$ is the required input voltage, L_c is the volume level chosen by the user, $\Delta L = L_c - L_{\text{ref}}$ is the difference between L_c and the reference volume level, and U_{ref} is the reference input voltage.

The reference level is measured under the same circumstances, and vary accordingly with the exact speakers.

Thus, the average power consumption of the speaker is

$$P_{\text{speaker}} = \frac{1}{t_0} \int_0^{t_0} \Re Z(f) \frac{U_{\text{in,req}}(f)^2}{|Z(f)|^2} dt \quad (25)$$

where f is the frequency (related to time t), and $Z(f) \in \mathbb{C}$ is the complex impedance of the speaker coil.

5 Battery Degradation

5.1 Overview

By Nernst equation, we have

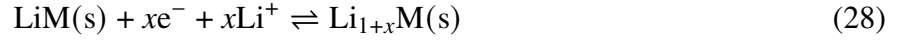
$$\mathcal{E} = \mathcal{E}^\circ - \frac{RT}{nF} \log Q \quad (26)$$

where $\mathcal{E}$ is the electromotive force, $\mathcal{E}^\circ$ is the standard electromotive force (magnitude 3.0401 V), R is the universal gas constant, T is the temperature in Kelvin, n is the number of moles of electrons transferred in the reaction, F is the Faraday constant, and Q is the reaction quotient.

Now we substitute $\mathcal{E} = U_{\min}$, where $U_{\min}$ is the minimum battery output voltage, and we have $Q = K$, the equilibrium constant. Thus,

$$U_{\min} = \mathcal{E}^\circ - \frac{RT}{nF} \log K \Rightarrow K = \exp \frac{nF(\mathcal{E}^\circ - U_{\min})}{RT} \quad (27)$$

We consider the total reaction



and the forward reaction (charging state), we have the equilibrium constant

$$K_+ = c^{-x}(\text{Li}^+) = \left(\frac{M(\text{Li}) \cdot V}{m(\text{Li}^+)} \right)^x \quad (29)$$

where V is the volume of the electrolyte. Hence,

$$m(\text{Li}^+) = M(\text{Li}) \cdot V \cdot K_+^{-\frac{1}{x}} = M(\text{Li}) \cdot V \cdot \exp - \frac{nF(\mathcal{E}_+^\circ - U_{\min+})}{xRT} \quad (30)$$

If we define $1 - r_+$ as the loss ratio of lithium ions in the electrolyte, then

$$r_+ = 1 - \frac{m(\text{Li}^+)}{m_0} = 1 - \frac{M(\text{Li}) \cdot V}{m_0} \cdot \exp - \frac{nF(\mathcal{E}_+^\circ - U_{\min+})}{xRT} \quad (31)$$

Now we consider the reverse reaction (discharging state), we have the equilibrium constant

$$K_- = c^x(\text{Li}^+) = \left(\frac{m(\text{Li}^+)}{M(\text{Li}) \cdot V} \right)^x \quad (32)$$

Using similar methods and note that $\mathcal{E}_+^\circ = -\mathcal{E}_-^\circ$, we have

$$r_- = 1 - \frac{M(\text{Li}^+) \cdot V}{m_0} \exp - \frac{nF(U_{\min-} + \mathcal{E}_+^\circ)}{xRT} \quad (33)$$

and the capacity after N cycles is (assume T is constant, and N not necessarily an integer)

$$C_N = C_0 r_+^N r_-^N \quad (34)$$

For non-complete draining, we may assume that

$$N = \sum_i p_i \quad (35)$$

where p_i is the percentage of capacity drained in the i -th cycle.

5.2 Internal Resistance

Citing the works from previous scholars, we may use a polynomial to model the internal resistance of the battery

$$r_i = \varphi_N \sum_{i=0}^4 A_{i,4-i}(\text{SOC})^i T^{4-i} \quad (36)$$

where $A_{i,4-i}$ are constants related to the specific battery, and $\varphi_N = \varphi_0^N$ (we assume) is a coefficient related to the number of cycles.

6 Peripherals

6.1 Camera

Suppose that the power consumption of the camera is proportional to the number of pixels and the frame rate, we have

$$P_{\text{camera}} = \gamma \cdot N_{\text{pixels}} \cdot f_{\text{frame}} + P_{\text{default}} \quad (37)$$

6.2 Microphone

Suppose that the power consumption of the microphone is proportional to the sampling rate and bit rate, we have

$$P_{\text{microphone}} = \delta \cdot f_{\text{sample}} b_{\text{sample}} + P_{\text{default}} \quad (38)$$

7 Heating Effects

For heating analysis, we consider the processing units as pure resistances. Thus, the heat generated by the units can be described as

$$q_{\text{P.U.}} = \frac{dQ_{\text{P.U.}}}{dt} = UI \quad (39)$$

$$q_{\text{battery}} = I^2 r \quad (40)$$

then we may use the Fourier's law to analyze the heat transfer. Suppose the effective distance between the heat source and the other components is d , and the effective area of heat transfer is A , then we have

$$q_{\text{disp}} = k_{\text{P.U.}} A_{\text{P.U.}} \frac{T_{\text{P.U.}} - T_{\text{env}}}{d} + k_{\text{battery}} A_{\text{battery}} \frac{T_{\text{battery}} - T_{\text{env}}}{d} \quad (41)$$

where k is the thermal conductivity of the material between the source and the overall phone environment, T_{source} is the temperature of the heat source, and T_{env} is the temperature of the phone environment, where all other components are in direct contact with.

Then, for the thermal sources, we have

$$q_{\text{P.U.}} - k_{\text{P.U.}} A_{\text{P.U.}} \frac{T_{\text{P.U.}} - T_{\text{env}}}{d} = c_{\text{P.U.}} m_{\text{P.U.}} \frac{dT_{\text{P.U.}}}{dt} \quad (42)$$

$$q_{\text{battery}} - k_{\text{battery}} A_{\text{battery}} \frac{T_{\text{battery}} - T_{\text{env}}}{d} = c_{\text{battery}} m_{\text{battery}} \frac{dT_{\text{battery}}}{dt} \quad (43)$$

where c is the specific heat capacity of the processing unit material, and m is the mass of the processing unit. For the overall phone environment, we have

$$q_{\text{disp}} - q_{\text{ext}} = c_{\text{env}} m_{\text{env}} \frac{dT_{\text{env}}}{dt} \quad (44)$$

where c_{env} is the overall heat capacity of the phone environment, and q_{ext} is the heat loss to the external environment, which can be modeled using Newton's law of cooling

$$q_{\text{ext}} = h A_{\text{phone}} \frac{d(T_{\text{env}} - T_{\text{ext}})}{dt} = h A_{\text{phone}} \frac{dT_{\text{env}}}{dt} \quad (45)$$

where h is the overall heat transfer coefficient, $A_{\text{phone}} = 2A_{\text{screen}} \approx A_{\text{P.U.}} + A_{\text{battery}}$ is the surface area of the phone, and T_{ext} is the temperature of the external environment, which we see as a constant.